



Geometry of the core of the triple helix of innovation game

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Abstract

The Triple Helix of innovation is a three-person cooperative game with transferable utility. The core is defined as the set of players' utilities that ensures individual and collective interests and the constraints on these utilities. Geometrically, the core of the Triple Helix game is a polygon with at most six faces, meaning that it may have less. The paper responds the question on how does the core form. Starting with the geometrical construction of the core of an imaginary Triple Helix game with six faces, we computed the coordinates of the 12 intersections of the six pairwise lines bounding the core and the ones of the vectors of the perimeter of the core. To illustrate, publications and patents data were collected from the Web of Science and the United States Patents and Trademark Office databases. The rules of the games were determined and the cores plotted. Results showed that the length of the faces of the core depends on bilateral and trilateral interactions only, and so does the surface area of the core as measurement of the synergy within the considered innovation system. Moreover, each bilateral collaboration engenders one face of the core and, with the trilateral collaboration, contributes a second; consequently, when that bilateral collaboration vanishes the core loses a face and the length of the second face decreases, so does the synergy within the system. The theory of the Triple Helix stating that the interactions among innovation actors generates the synergy that leads to innovation is then confirmed. The more an interaction, the lengthier the side of the core it engenders.

Keywords Triple Helix · Innovation system · Synergy · Game theory · Core

Introduction

The Triple Helix of innovation (Etzkowitz & Leydesdorff, 1995) as a complex system (cf. Katz, 2006, 2016 for an innovation system in general; and Leydesdorff, 2003 for the Triple Helix innovation system in particular) is a three-person cooperative game with transferable utility (Mègnigbêto, 2018c). The core introduced as a solution concept to the cooperative game with transferable utility by Gillies (1953) is considered as a measurement of the synergy within a Triple Helix innovation system (Mègnigbêto, 2018c). The core of a

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cooperative game is defined as the set of actions no individual player has an incentive to deviate from; it is so that it leaves “no coalition in a position to improve the payoffs of all its members” (Shapley & Shubik, 1973, p. 40). In the Triple Helix study, the core may be interpreted as the range of all possible compromises players could reach based on their interests and the constraints on these interests, with the target of creating synergy (Mêgnigbêto, 2018c, p. 1123).

Many studies used the theory of the Triple Helix to assess diverse economies in the world; some sought to measure the synergy within the economy using units of analysis such as publication (e.g. Leydesdorff, 2003), patent (Patra & Muchie, 2018) or technological cooperation (Ivanova et al., 2014). A few resort to game theory for the same purpose (among them Acuña et al., 2024; and Mêgnigbêto, 2018c for example); the core is used as measurement of the synergy. The existence of the core of the innovation game determines the one of the synergy within the innovation system; the surface area of the core measures the extent of the synergy. “The core of the Triple Helix game is determined by the output percentage share of each act or on its own (the lower bound) and the total output percentage share the actor produced within the system, included in collaboration with other actors (the upper bound) (Mêgnigbêto, 2018c, 2024a). The core is then the intersection of maximum and minimum of actor’s strategic interests; any variation within these interests has influence on the borders and the surface area of the core. Therefore, understanding how the core forms helps in understand how the synergy occurs; this motivates the study of the geometry of the core.

The core of a cooperative game with transferable utility may be studied from a geometrical point of view. The analysis of various Triple Helix innovation systems game (cf. Mêgnigbêto, 2018b, 2018c, 2019, 2024b, 2024a) revealed that the form and position of the core depend on the characteristics of the game (see also Acuña et al., 2024). They revealed that the core may be a polygon of which number of faces varies with the configuration of the game: “each time a bilateral relationship is missed, the core loses a side and the synergy [within the innovation system] decreases” (cf. Mêgnigbêto, 2024a). Derk and Kuipers (2002) demonstrated, following Gillies (1953), that if the number of players is n , the core is a polyhedral with at most $n!$ extreme points. Shapley (1965, 1971) particularly studied the geometry of the core of convex games; he found that the convexity shapes its form: if n is the number of players, the dimension of the core is at most $n - 1$ if the game is convex and exactly $n - 1$ if the game is strictly convex; if the game is strictly convex, the core has exactly $2^n - 2$ polyhedral faces of dimension $n - 2$ (see also González-Díaz & Sánchez-Rodríguez, 2008, p. 103 ff.) and exactly $n!$ extreme points (Derk & Kuipers, 2002).

Because the Triple Helix of innovation game has three players (Mêgnigbêto, 2018c) and the game is convex anyway (and may be strictly convex, Mêgnigbêto, 2024a), it comes that its core is a polygon of at most 6 faces. The number of faces (and hence the number of vertices) equals 6 if the game is strictly convex; the core is then a hexagon (see also Bloch & de Clippel, 2010, p. 5 who stated that “if the core has the shape of a polygon, it is an hexagon characterised by six extreme points”). The core is not, however, a regular polygon because the length of its faces varies; when publication is taken as unit of analysis, the width of the core is determined by the contribution of the actor industry and the length by the contributions of both university and government (cf. Mêgnigbêto, 2018b, 2018c, 2019).

The studies above did not deal with how does the core of a Triple Helix game form, how do the utilities (minimum or maximum) of each actor contribute to the core or what faces depend on what bilateral collaboration. In other words, they did not interest in the geometry of the core of the Triple Helix of innovation game. Mêgnigbêto (2018b, 2018c, 2019) stated that the amplitude of each actor’s utility as given by the analytical form of the core

shows that actor's contribution to the formation of the core; he also noticed, using publication as unit of analysis, that the width of the core is determined by the actor Industry's contribution and the length by the contributions of actors University and Government. The unique research question this paper answers is formulated as follows: how do the Triple Helix game players' contributions shape the core?

Methods and object

We start by drawing the core of an imaginary Triple Helix of innovation game, consisting in tracing three pairwise parallel lines. Next, we determine the ternary coordinates of the intersections of the six lines and the ones of the vectors from the perimeter of the core. Prior studies used publications data only to determine the Triple Helix core; in this paper, we couple publications and patents data to detect any change in the formation of the core; so, data for the application of the results of the analyses are collected from the Web of science for publications and the United States Patents and Trademark Office for patents over the decade 2011–2020. The search strategy used was developed and tested by Leydesdorff (2003, p. 458) and Park et al., (2005, p. 13 ff) while searching the Web of Science and adopted by Ye et al. (2013) and Mênigbêto (2015b, 2018a); it consisted in attributing records to Triple Helix actors on the basis of keywords¹ in eleven steps.² We adapted it to patents search.³

The collected data were entered into a worksheet to compute Triple Helix spheres data, the characteristics functions, the minimum and maximum interests of actors for the games under consideration. Then, we used the R statistical software (R Development Core Team, 2023) and its packages “CoopGame” (Staudacher & Anwander, 2019) to determine the vertices of the cores (with their ternary coordinates), “Ternary” (Smith et al., 2023) to convert ternary coordinates to Cartesian coordinates with the objective of drawing the cores and computing its surface area. The different plots of the core are obtained either on a ternary diagram with TernaryPlot.com (Blom, 2023) – to show the global environment – or on a Cartesian plan with GeoGebra (GeoGebra GmbH, 2023; Hohenwarter, 2002) – to show the local environment.

¹ In Web of Science, the presence of one of the strings “UNIV, COLL, ECOLE” in an address attributes the record to University, the presence of one of the strings “GMBH, CORP, LTD, AG” in an address attributes the record to Industry and the presence of one of the strings “NATL, NACL, NAZL, GOVT, MINIST, ACAD, NIH” in an address attributes the record to Government.

² (1) selection of all the scientific output—(2) selection of all university output—(3) selection of all industry output—(4) selection of all governmental output—(5) selection of output government and university share—(6) selection of output government and industry share—(7) selection of governmental output only—(8) selection of output university and industry collaboration output—(9) selection of university and government collaboration output—(10) selection of industry and government collaboration output—(11) selection of university, industry and government collaboration output.

³ In USPTO databases, the string are “university, universidade, school, ecole, college” for University, “indus, indústria, corp or corporation, co., ltd, limited, gmbh, inc., lld, llc, a/s, enterpr, entrepr, societ, group, sarl, sa, holding, compa” for Industry and “commissariat, national, hospital, hospital, minist, agenc, center, centre, nih, nhs, governo, bundes, oveheid, acad, insttit” for Government.

Analyses

Vertices of the core

From a geometrical point of view, the core of a three-person cooperative game with transferable is the surface area of the ternary triangle delimited by the six pairwise parallel lines that determine the lower and the upper values of players interests (Mêgnigbêto, 2018c, p. 1130). Fig. 1 shows the geometrical closer environment of the core of an imaginary Triple Helix game; the core is a hexagon. The six pairwise parallel lines are marked with their equation on a ternary diagram (i.e., U, I or G followed by the subscript “min” or “max” for “minimum” or “maximum”: $(U_{\min})$, $(U_{\max})$, $(I_{\min})$, $(I_{\max})$, $(G_{\min})$ and $(G_{\max})$). See System of Eqs. (1):

$$\left\{ \begin{array}{l} U_{\min} = U \\ I_{\min} = I \\ G_{\min} = G \\ U_{\max} = U_0 = U + UI + UG + UIG \\ I_{\max} = I_0 = I + UI + IG + UIG \\ G_{\max} = G_0 = G + UG + IG + UIG \end{array} \right. \quad (1)$$

where U, I, G represent the output percentage share of university, industry and government on their own respectively; UI, UG, IG the output percentage shares of university and industry, university and government, industry and government shared respectively (i.e., the

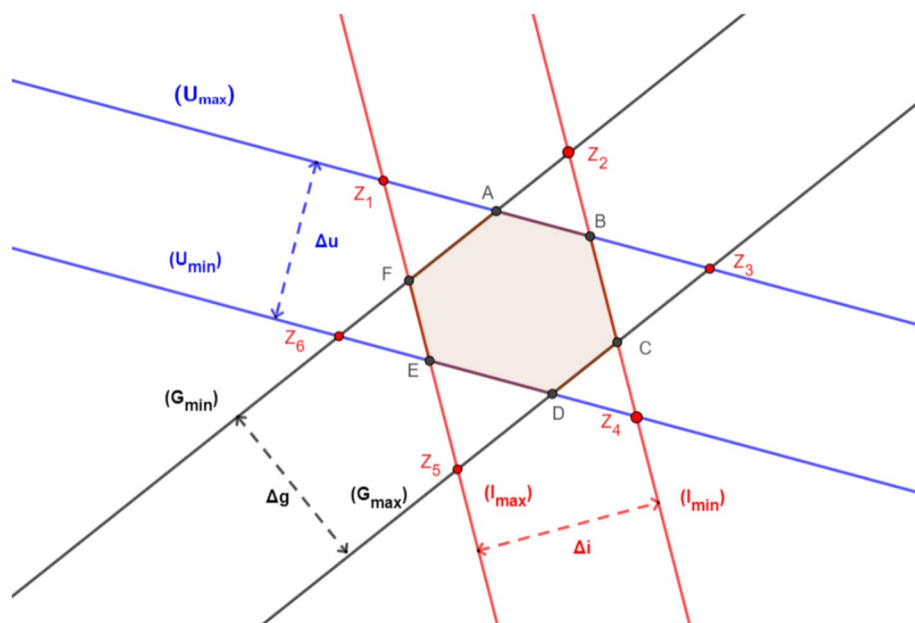


Fig. 1 Local environment of the core of an imaginary Triple Helix of innovation game

bilateral collaboration outputs), and UIG the output percentage share (i.e., the trilateral collaboration output).

The six pairwise parallel lines have 12 intersections of which six have their coordinates out of range (we named them Z_1 to Z_6) and six are the vertices of the core (we named them A, B, C, D, E, F). Globally, the intersection of two lines both representing a minimum or a maximum value is out of range; and, the intersection of lines, the one representing a minimum value and the other a maximum value constitutes a vertex of the core (cf. Table 1). Fig. 1, Δ_u , Δ_i and Δ_g are the distances between two parallel lines; there are also the amplitude – i.e., the difference between the maximum and the minimum – of the utilities of players as indicated in System of Eqs. (2):

$$\begin{cases} \Delta_u = U_{\min} - U_{\max} = UI + UG + UIG \\ \Delta_i = I_{\min} - I_{\max} = UI + IG + UIG \\ \Delta_g = G_{\min} - G_{\max} = UG + IG + UIG \end{cases} \quad (2)$$

In Table 1, the coordinates of the intersections of the six pairwise parallel lines are determined as follows: the point at the intersection of two lines has two predefined coordinates, each from each line; for example, the intersection of the lines ($U_{\max}$) and ($I_{\min}$) has its University axis coordinate equal to $U_{\max}$ and its Industry axis coordinate equal to $I_{\min}$. Because, the three coordinates should sum up to $T (= 100)$, the third coordinate – i.e., the Government axis one – is derived and equals $T - U_{\max} - I_{\min} = (U + I + G + UI + UG + IG + UIG) - (U + UI + UG + UIG) - I = G + IG$; this intersection is the point A. When at least one coordinate is higher than the maximin value or lower than the minimum value following an axis, the resulting point is out of the core; this is the case of the point Z_1 , for which the Industry axis coordinate is $I - UG - UIG$, which is lower than I , the minimum value of the utilities of actor Industry.

Vectors of the perimeter of the core

Table 2 displays the ternary coordinates of the vectors based on the core's vertices. On each axis, the coordinate of a vector is the difference between the ones on the same axis of its extreme points. The table informs on how the core forms and, particularly, which Triple Helix relationship each side of the core stems from. As an example, $\overline{AB}$ has a null coordinate on the University axis, IG coordinate on the Industry axis and the opposite on the Government axis, meaning that the side AB of the core originates from the bilateral collaboration between actors i and g. The null value as coordinate justifies that the corresponding vector is parallel to the associated axis; for example, $\overline{AB}$ is parallel to the University axis. With the same logical reasoning, the side CD stems from the bilateral collaboration between actors u and g, and the side EF from the bilateral collaboration between actors u and i; however, the sides BC, DE and AF add to each bilateral collaboration the trilateral one. In conclusion, the length of the faces of the core, and hence its surface area, depend on the bilateral and trilateral collaborations and not on the actors' individual outputs. This result was predictable from System (1) which shows that the core's local environment does not depend on anything else than the bilateral and trilateral collaborations. Further considerations can then be done.

Let us suppose that the industry-government collaboration is missing (i.e., $IG=0$); then, following row 1 Table 2, the points A and B coincide and following row 4 the length of the face DE is reduced to the quantity UIG; moreover, the rows 2 and 11 Table 1 are

Table 1 Intersections of lines delimiting the core of a Triple Helix game

Row	Intersection of lines	Ternary coordinates			Point (Observations)
		University axis	Industry axis	Government axis	
1	$(I_{\min})$ and $(U_{\min})$	U	I	G + UI + UG + IG + UIG	Z_6 (Government axis coordinate is out of range)
2	$(I_{\min})$ and $(U_{\max})$	U + UI + UG + UIG	I	G + IG	A
3	$(I_{\min})$ and $(G_{\min})$	U + UI + UG + IG + UIG	I	G	Z_2 (University axis coordinate is out of range)
4	$(I_{\min})$ and $(G_{\max})$	U + UI	I	G + UG + IG + UIG	F
5	$(I_{\max})$ and $(U_{\min})$	U	I + UI + IG + UIG	G + UG	D
6	$(I_{\max})$ and $(U_{\max})$	U + UI + UG + UIG	I + UI + IG + UIG	G - UI - UIG	Z_3 (Government axis coordinate is out of range)
7	$(I_{\max})$ and $(G_{\min})$	U + UG	I + UI + IG + UIG	G	C
8	$(I_{\max})$ and $(G_{\max})$	U - UG - IG - UIG	I + UG + IG + UIG	G + UG + IG + UIG	Z_5 (University axis coordinate is out of range)
9	$(U_{\min})$ and $(G_{\min})$	U	I + UI + UG + IG + UIG	G	Z_4 (Industry axis coordinate is out of range)
10	$(U_{\min})$ and $(G_{\max})$	U	I + UI	G + UG + IG + UIG	E
11	$(U_{\max})$ and $(G_{\min})$	U + UI + UG + UIG	I + IG	G	B
12	$(U_{\max})$ and $(G_{\max})$	U + UI + UG + UIG	I - UG - UIG	G + UG + IG + UIG	Z_1 (Industry axis coordinate is out of range)

Coloured rows identify vertices of the core

Table 2 Ternary coordinates of vectors in the core of a Triple Helix game

Row	Vector	U axis	I axis	G axis
1	$\overline{AB}$	0	IG	− IG
2	$\overline{BC}$	UI + UIG	− UI − UIG	0
3	$\overline{CD}$	− UG	0	UG
4	$\overline{DE}$	0	− IG − UIG	IG + UIG
5	$\overline{EF}$	UI	− UI	0
6	$\overline{FA}$	UG + UIG	0	− UG − UIG

equivalent, and the points A and B (vertices of the core, and also Z_2 outside of the core) coincide; it was like Fig. 1, the line ($U_{\max}$) moved upwards so that A and B coincide with Z_2 . Similarly, if the university-government collaboration vanishes (i.e., $UG=0$), the rows 5 and 7 Table 1 are equivalent; then, the points C and D (vertices of the core, and also Z_4 outside of the core) coincide, and the length of the side AF reduces to UIG; and if the university-industry collaboration is null (i.e., $UI=0$), the rows 4 and 10 Table 1 are equivalent, and the points E and F (vertices of the core, and also Z_6 outside the core) coincide and the length of the side BC reduces to UIG. We can conclude, then, that each bilateral collaboration within the Triple Helix system engenders a face of the core and contribute a second face; consequently, when a bilateral collaboration is missed, the face of the core it engenders disappears and the one it contributes diminishes in length.

Illustrations

In this section, we analyse publications and patents data of the World breakdown by Triple Helix sphere (Table 3). Publications data show that industry is the lowest producer, university the top one and government as the intermediate; patents data rank actors differently: industry is at the top, followed by university and government is at the rear.

Let us note u , i and g the three players of the Triple innovation game and u , i , g , ui , ug , ig and uig the coalitions they may form. The characteristic functions of the associated Triple Helix games Table 4 inform that for publications data, coalitions player u is member of are ranked first (ug , ui , u in this order) followed by ig , g and i ; and, that for patents data, coalitions player i is member of are ranked first (ui , ig , i in this order), followed by ug , u and g .

Figure. 2 and Figure. 3 represent the cores of the two games. Their surface area (as percentage of the surface of the equilateral triangle) is low cf. Table 5, making the cores

Table 3 Publications and patents data of the World (2011–2020) per Triple Helix spheres

		U	I	G	UI	UG	IG	UIG	Total
Publications	#	19,980,635	308,252	682,390	820,536	1,599,156	122,496	108,062	23,721,527
	%	84.587	1.305	2.889	3.474	6.770	0.519	0.457	84.587
Patents	#	193,475	3,417,450	40,960	45,797	2,475	4,089	122	3,704,368
	%	5.223	92.255	1.106	1.236	0.067	0.110	0.003	100

Table 4 Characteristic functions of the World publications and patents Triple Helix games

	u	i	g	ui	ug	ig	uig
Publications	84.587	1.305	2.889	89.365	94.245	4.712	100
Patents	5.223	92.255	1.106	98.714	6.395	93.471	100

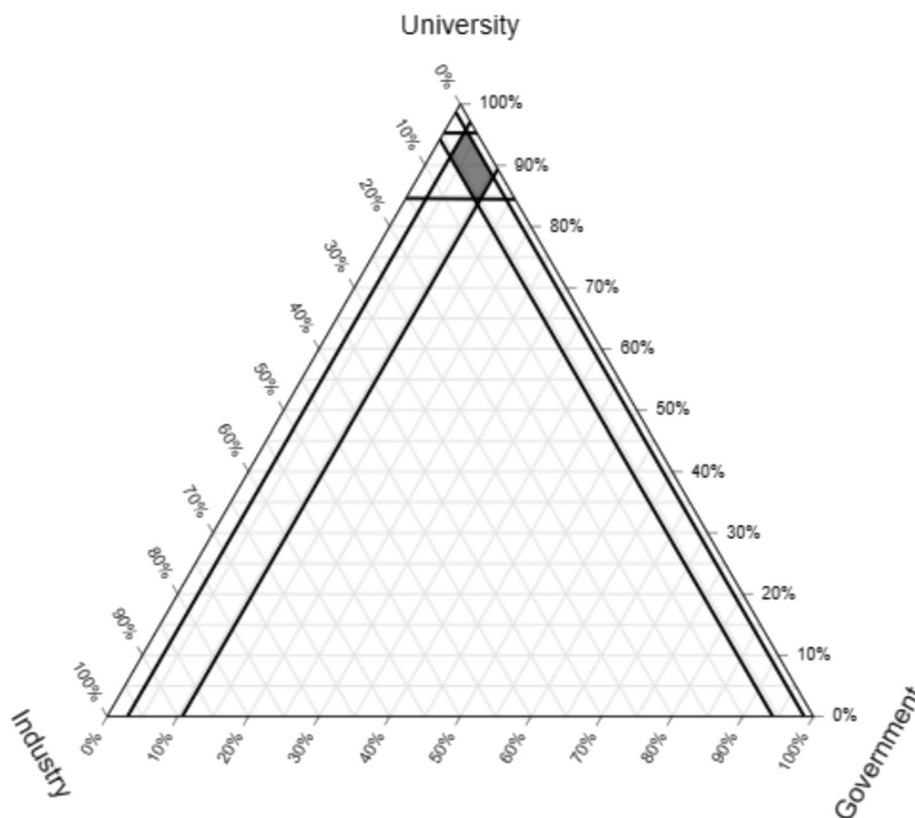


Fig. 2 Core of the World publications Triple Helix game (2011–2020): The diagram is designed using TernaryPlot.com (Blom, 2023). The core is the coloured area on the triangle

almost “invisible” on the supporting triangle or their form somewhat undistinguishable; so, we present below a “zoomed version” of the cores on a Cartesian plan Fig. 4 and Fig. 5 plotted with Geogebra (GeoGebra GmbH, 2023; Hohenwarter, 2002; Hohenwarter et al., 2018). Supplementary, Fig. 4 shows that the transforming of ternary coordinates into Cartesian ones operates a vertical axis symmetry on the core plotted on a ternary diagram, so that on the ternary diagram, the points to the left appear to the right on the Cartesian representation, and vice versa.

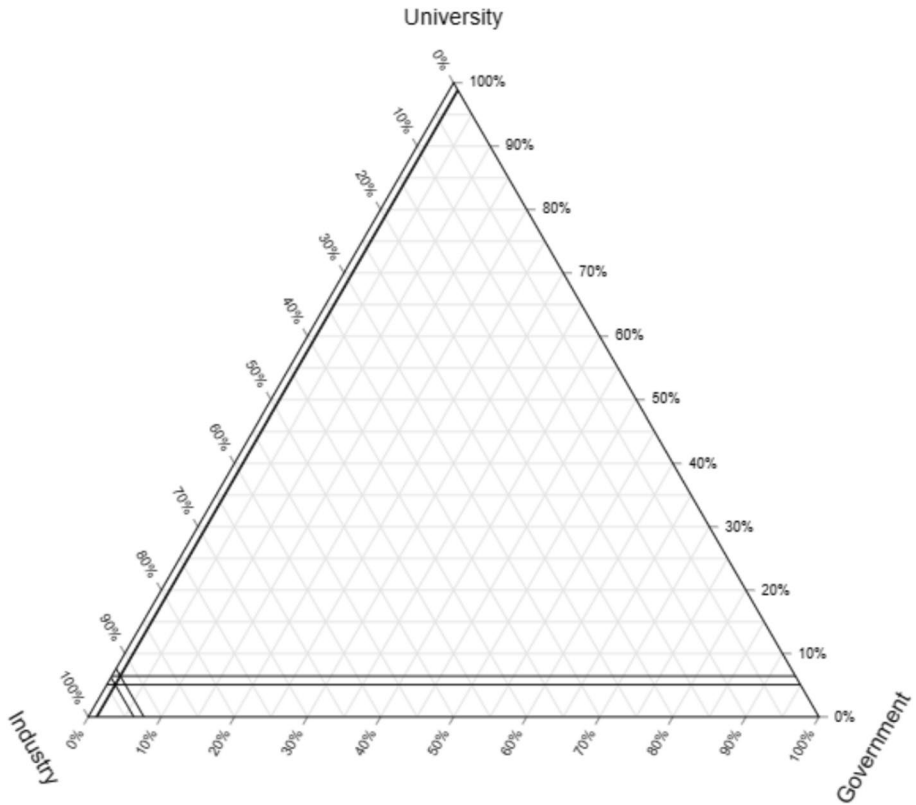


Fig. 3 Core of the World patents Triple Helix game (2011–2020) The diagram is designed using Ternary-Plot.com (Blom, 2023). The core is the coloured area on the triangle. It is almost invisible

Table 5 Analytic form and surface areas of the cores

Country	Publications	Patents
Analytic form	$\begin{cases} 84.59 \leq x_u \leq 95.29 \\ 1.30 \leq x_i \leq 5.75 \\ 2.89 \leq x_g \leq 10.63 \\ x_u + x_i + x_g = 100 \end{cases}$	$\begin{cases} 5.223 \leq x_u \leq 6.529 \\ 92.255 \leq x_i \leq 93.605 \\ 1.106 \leq x_g \leq 1.286 \\ x_u + x_i + x_g = 100 \end{cases}$
Surface area ($\times 1000$)	729.263	4.608

Discussion

Interactions shape the core

The core of the Triple Helix of innovation game results from the intersections of the six pairwise lines bounding players interests; the faces' length and the surface area of the core depend on the interactions among innovations actors and on nothing else. So, when a bilateral collaboration vanished, one face of the core disappears, the length of a second face decreases and the surface area diminishes; however, when the trilateral collaboration does

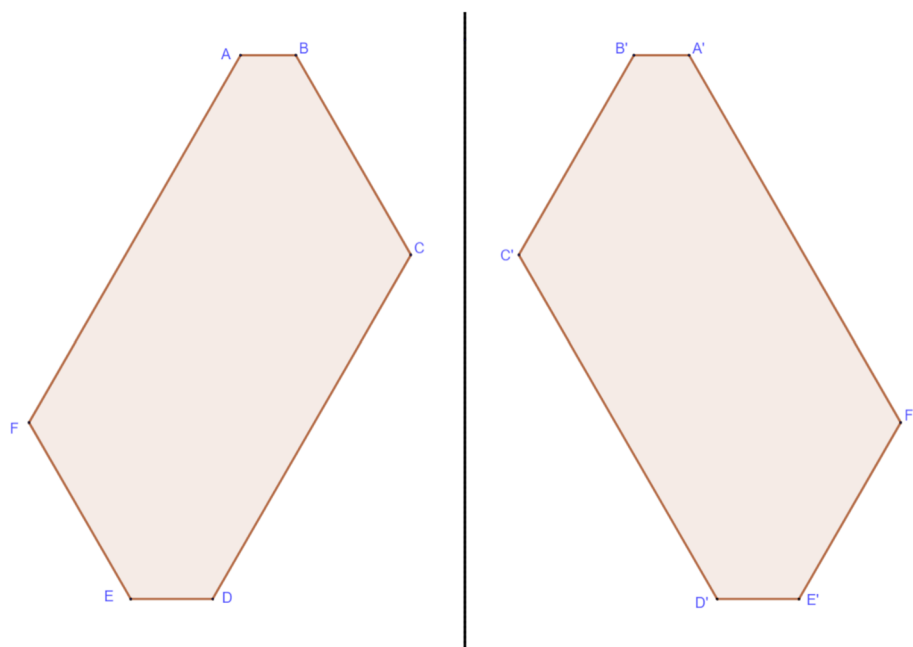
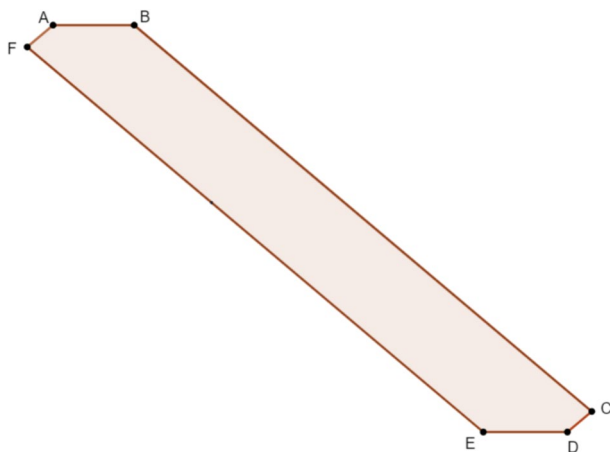


Fig. 4 Core—of the World publications game—on a ternary diagram (to the right) as the vertical axial symmetry of core on a Cartesian plan (to the left)

Fig. 5 Core of the World patents Triple Helix game (on a Cartesian plan)



not exist, no face disappears, because no face length depends on the extent of the trilateral collaboration only. Mênigbêto (2019, 2024a) recorded such cases even though no explanation was provided. Consequently, the synergy within a Triple Helix innovation system measured with the core only depends on the interactions between innovators and on anything else. We can, therefore, conclude that the interactions among innovators shape the core. This finding is conform to the Triple Helix theory stating that the !!!synergy within the Triple Helix is the outcome of the interactions and the overlay of communication

between actors (by the means of collaboration), and feeds innovation (Etzkowitz & Leydesdorff, 1995, 2000; Leydesdorff & Etzkowitz, 2001).

This result does not imply that actors' own outputs have no incidence on the core; indeed, the minimum and the maximum values (as in the analytic form of the core) delimit the parts of the ternary diagram out of the core and, hence, the one in the core, and as a result, the position of the core on the ternary diagram which varies. Mênigbêto (2018b, 2019) distinguished i) the university-dominated core, where actor University has the highest minimum and maximum values so the core positions close to the University apex (see also Acuña et al., 2024), ii) the government-dominated core, where actor Government has the highest minimum and maximum values so the core positions close to the Government apex, and, iii) the balanced core, where the shares of university and government are balanced and the core is located at the middle of the side of the triangle opposite to the Industry apex. Let us recall that publication was taken as unit of analysis for this categorisation; because industry has a negligible share as compared to the two other actors, it does not appear. The more an actor individually produces, the more he does in collaboration; therefore, for publications data, actors own and total production rank u , g and i in this order; bilateral collaborations involving player u are higher than the ones involving g , and as a result $UG > UI > IG$ (cf. e.g., data treated in the following studies: Leydesdorff, 2003; Mênigbêto, 2013, 2014a, 2015a, 2016). On the other hand, actors own and total patents production rank i , u and g in this order; bilateral collaborations i involves in are higher than the one u involves in; so, the ranking $UI > IG > UG$ is expected and registered. The highest share an actor has then affects roles within the set of players. As an illustration, the marginal contribution of a player (in the case of the Triple Helix of innovation game $U_{\max}$, $I_{\max}$ and $G_{\max}$ for players u , i and g respectively) reflects the “power” of that player in the game; it appears like the threat a player exerts on its partners while negotiating coalitions, or the power of negotiation of a player (Stolwijk, 2010). Therefore, each player used its “power” to lead to and dominate coalitions and, hence, payoffs sharing. The position of the core on a ternary diagram reflects it: the core of patents game is close to the Industry axis (to the left) and the one of the publications data is close to the University axis (to the top) as in Fig. 2 and Fig. 3. Overall, when an actor has the largest marginal contribution (and perhaps, the largest individual output), the core is close to its apex.

Width and length of the core rely on ranking of players

The core of the Triple Helix game is an irregular hexagon; by definition, this geometrical form has no width, length or height; therefore, we will consider the height and the basis of the parallelogram containing it as the width and the length of the core. Let us consider the parallelogram AZ_3DZ_6 in Fig. 6 (or Z_1BZ_4E in Fig. 7); it lays on the lengthiest faces of the core and contains the core. In Fig. 6, Δ_i represents the width of the parallelogram (and hence, of the core), but the length results both from Δ_u and Δ_g ; this finding is conform to the one by (Mênigbêto 2018b, 2018c, 2019) stating that the width of the core is determined by the variation of the utilities of actor i . In Fig. 7, however, the width equals Δ_g and the length results from both Δ_u and Δ_i contradicting the previous result. Note that this figure displays the core of a Triple Helix game with patents data and, patents is mainly produced by industry. To understand how this occurs, some considerations on the geometry of the core is needed.

Let us observe the position of the core as reference to the vertical and dotted line passing through the point A in both Fig. 6 and Fig. 7. We remark that the lower part formed by

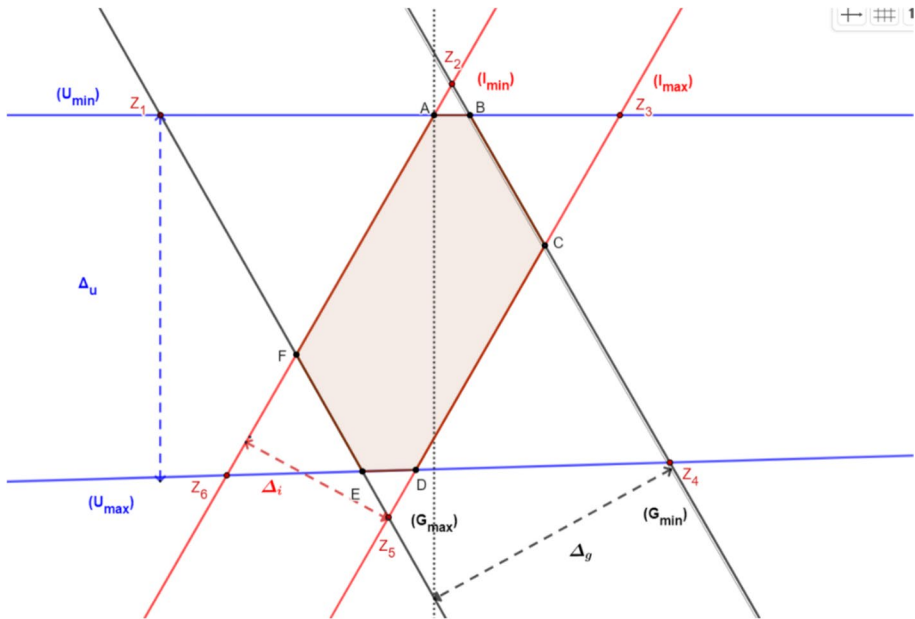


Fig. 6 Core of the World publications Triple Helix game

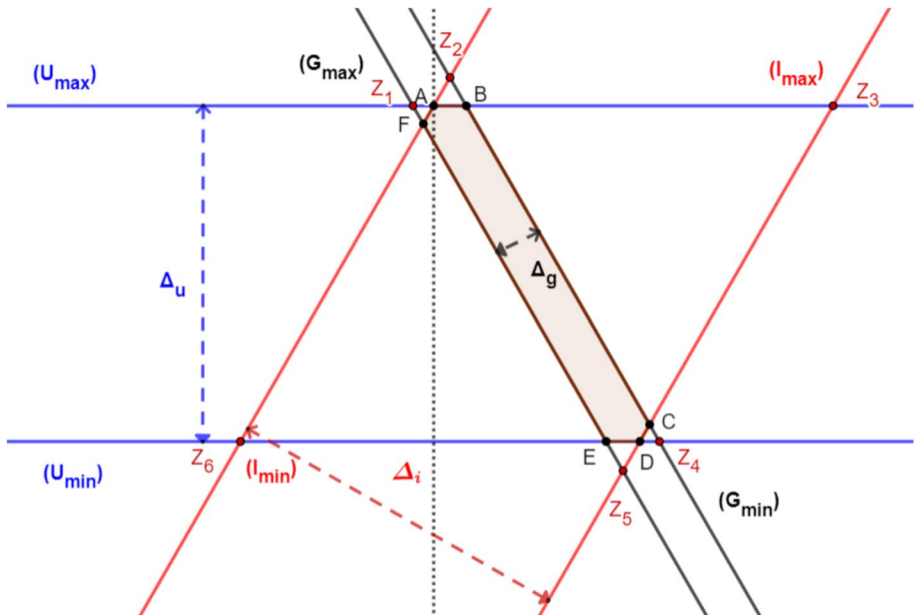
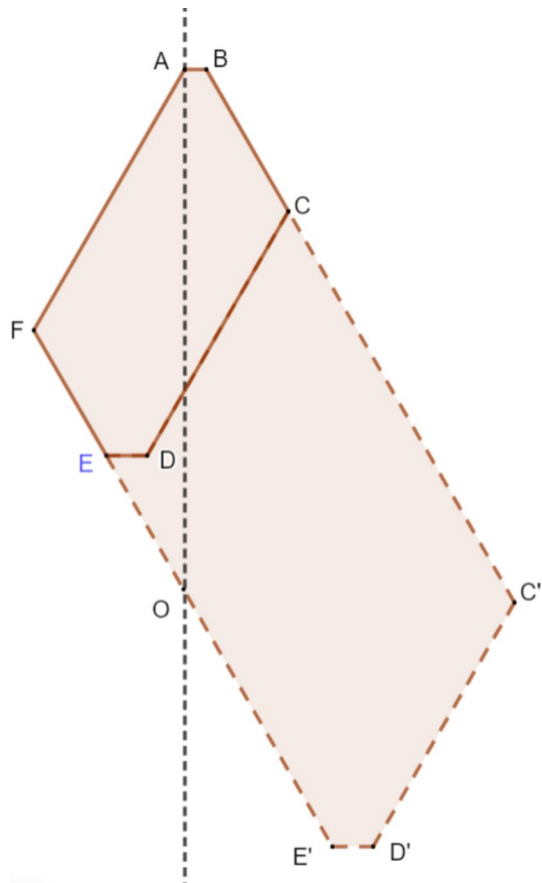


Fig. 7 Core of the World patents Triple Helix game in its local environment

the triangle DEF deviates to the left in Fig. 6 whereas in Fig. 7, the lower part formed by the triangle CDE deviates to the right. How could this happen? According to our findings, faces CD and AF are parallel; they are engendered by the quantities UG and $UG + UIG$ respectively; faces EF and BC are parallel too and engendered by the quantities UI and $UI + UIG$ respectively; and faces AB and ED are parallel and engendered by the quantities IG and $IG + UI$ respectively. Let us suppose that the quantity UI increases to be higher than the quantity UG; it means that in Fig. 6, the face EF increases to be lengthier than CD. The polygon ABCDEF becomes $ABC'D'E'F'$; it has changed then its positioning towards the vertical dotted line: ED moves to the right. Such a positioning recalls the one of the core in Fig. 7 (of the game with patent as unit of analysis). So, the faces AF on the line (I_{max}) and CD on the line (I_{min}) bound the width of the core, which becomes Δ_g . The translation the points C, D and E undergo to become C' , D' and E' respectively, is visible on Fig. 8. The same reasoning may be done on the reverse direction considering Fig. 7, the quantity UI decreasing and, consequently, the face ED changing to be lengthier than EF.

Fig. 8 Relative positions of the cores of World publications (ABCDEF) and patents ($ABC'D'E'F'$)



Conclusion

The objective of this paper was to examine how does the core of the Triple Helix of innovation game form; more precisely, how the players' own and collective outputs influence the formation of the core and hence the existence and the extent of the synergy. Having started by drawing the core of an imaginary Triple Helix game, we determined the coordinates of its vertices and the ones of the vectors of its perimeter; for illustration purpose, we collected and treated publications and patents data from the Web of Science and the United States Patents Trademark Office databases. Results showed that the length of the faces of the core depends on bilateral and trilateral interactions only, and thus, does the surface area of the core as measurement of the synergy within the considered innovation system. Moreover, each bilateral collaboration engenders a face of the core and, with the trilateral collaboration, contributes a second. Consequently, when the bilateral collaboration vanishes, the core loses its face stemming from that collaboration, and the length of the second face decreases, so the synergy within the system diminishes. The theory of the Triple Helix stating that the interactions among innovation actors generates the synergy that leads to innovation is then confirmed. The more an interaction, the lengthier the side of the core it stems from.

This finding contributes to the choice and the use of an indicator to measure the synergy within a Triple Helix innovation system. Indeed, Leydesdorff (2003) stated that the interactions between the Triple Helix innovation actors generate probabilistic entropy and may be measured as an indicator of entropy using the Shannon's (1948) information theory; he then proposed the mutual information of which mathematical formulation was given by Hu et al. (2021). Krippendorff (2009a, 2009b) advised not to use this indicator; he argued that the mutual information lacks to measure the interactions between three variables because when it equals 0, it does not mean the absence of interaction (examples can be found in the following: Krippendorff, 2009b; Mégnigbêto, 2014b, 2016). Indeed, if the synergy is created by interactions, an increase in the collaborative data should return in an increase in the mutual information, which means an increase in the synergy within the system. However, the mutual information does not behave like that: it reaches its maximum value in the case of equiprobability and starts decreasing above due to the logarithmic function (Cover & Thomas, 2006; Yeung, 2008). Our finding indicates that any increase in collaborative outputs yields an increase in the extent of the core, which is in accordance with the Triple Helix theory. As an illustration, for the data we analysed, the mutual information equals -23.393 mbits for the publication system and -38.779 mbits for the patent system, meaning that the patent system has more synergy than that of publication. However, both bilateral and trilateral collaboration data are higher in the case of publication than the one of patent; as a result, one should expect the publication system be more synergetic than the patent one. Whereas any variation in any data (including individual output) influences the mutual information, only changes in collaborative data have incidence on the core. That is what our paper demonstrated.

The core of the Triple Helix of innovation game is the set of players interests and constraints on these interests (Mégnigbêto, 2018b, 2018c); geometrically, it is a hexagon ABC-DEF of which faces are pairwise parallel (AB and ED, CD and AF, BC and EF); in each couple of parallel faces, one is lengthier than the other ($AB > ED$, $CD > AF$ and $EF > BC$). These properties give the pattern of the core. So, due to the preponderance of actors' production, the orientation of the core changes. For publications, the core orients from top right to bottom left, while the core of patent game orients from top left to bottom right.

This result suggests that with another unit of analysis, another pattern of the core may display. The results may allow for the revision of the statement by (Mêgnigbeto 2018c, 2018b, 2019): the lowest producer's output determines the width of the core and the two major producers its length. Whatever the unit of analysis, the form of the core will still a polygon, but its dimensions will vary depends on the data and the actors. Another pattern of the core may be revealed by further studies, for example funding game where government is often at the top of the ranking.

Collaboration is recognized essential to knowledge production, sharing and transformation into innovation. Our findings stress the importance of collaboration in synergy creating in an innovation system. “Innovation is central to improvements in living standards” (OCDE & EUROSTAT, 2019, p. 19) which is the goal of all development policies, at local, national or regional level. Therefore, innovation policies should encourage and support collaboration between economy actors; this may be through the identification of priority areas of research and knowledge producing, grants and taxes policies.

The limitations of this study related to the data treatment. Indeed, records assignment to Triple Helix actors were based on keywords based on the literature. However, some records might have been wrongly assigned. Secondly, over the more than 27 million records from the Web of Science, 87% were categorised and over the around 7 million records from USPTO databases, 53% were assigned a category. A refining of keywords may help in improving this score.

Author contributions The author conceived this study, analysed data, presented and discussed the results.

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Data availability Data analysed in this study are collected form Web of Science and USPTO databases by the author; they are available on request.

Declarations

Conflict of interest There is no competing interest.

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